

# How do multi-agent LLM systems coordinate and communicate?

## Executive Summary

The coordination and communication of multi-agent Large Language Model (LLM) systems is a complex and multifaceted topic. Recent research has highlighted the importance of architectural structures, communication mechanisms, and evaluation frameworks in governing the behavior of these systems. This report provides an overview of the current state of knowledge in this area, including key findings, points of disagreement, and research gaps.

## Key Findings

Multi-agent LLM frameworks impose distinct architectural structures that govern how agents interact, store information, and coordinate tasks, as noted in [Understanding Multi-Agent LLM Frameworks: A Unified Benchmark and Experimental Analysis](#). Communication is an effective mechanism for coordinating the behaviors of multiple agents, broadening their views of the environment, and supporting their collaborations, as highlighted in [A Survey of Multi-Agent Deep Reinforcement Learning with Communication](#). The design and evaluation of multi-agent collaboration protocols remain significant challenges, especially for enterprise applications, as emphasized in [Towards Effective GenAI Multi-Agent Collaboration: Design and Evaluation for Enterprise Applications](#).

Specific techniques, such as Cayley graph optimization and interference-aware K-step reachable communication, have been proposed to improve the performance of multi-agent LLM systems. For example, [Cayley Graph Optimization for Scalable Multi-Agent Communication Topologies](#) proposes a family of circulant Cayley graphs whose generator sets are optimized to minimize diameter, directly targeting worst-case information propagation speed. Meanwhile, [Interference-Aware K-Step Reachable Communication in Multi-Agent Reinforcement Learning](#) introduces a novel communication protocol that enables agents to select collaborators under uncertainty, lacking a priori knowledge of which partners can deliver task-critical information.

## Points of Disagreement

There are some points of disagreement or tension between sources, particularly with regards to the importance of conventions and zero-shot coordination in multi-agent systems. For example, [Augmenting the action space with conventions to improve multi-agent cooperation in Hanabi](#) emphasizes the role of conventions in improving cooperation, while [Towards Zero-Shot Coordination between Teams of Agents: The N-XPlay Framework](#) focuses on the need for zero-shot coordination in complex, real-world scenarios. Further research is needed to resolve these disagreements and develop a more comprehensive understanding of the role of conventions and zero-shot coordination in multi-agent LLM systems.

## Research Gaps

Despite the advances in multi-agent LLM systems, there are still significant gaps in our understanding of these systems, particularly in terms of their coordination and communication. For example, [AEMA: Verifiable](#)

[Evaluation Framework for Trustworthy and Controlled Agentic LLM Systems](#) notes that evaluating LLM-based multi-agent systems remains a critical challenge, as these systems must exhibit reliable coordination, transparent decision-making, and verifiable performance across evolving tasks. Additionally, [Mapping Human Anti-collusion Mechanisms to Multi-agent AI Systems](#) highlights the need to adapt human anti-collusion mechanisms to AI settings, as multi-agent AI systems become increasingly autonomous and may develop collusive strategies similar to those observed in human markets and institutions.

Furthermore, there is a lack of comprehensive evaluation frameworks for multi-agent LLM systems, as well as limited understanding of how to adapt human anti-collusion mechanisms to AI settings. Insufficient exploration of the role of conventions and zero-shot coordination in complex, real-world scenarios is also a significant research gap. Addressing these gaps will require further research and development of new methodologies and frameworks for evaluating and improving the performance of multi-agent LLM systems.

## Conclusion

In conclusion, the coordination and communication of multi-agent LLM systems is a complex and multifaceted topic, with various approaches, techniques, and challenges being explored in the field. While significant advances have been made in recent years, there are still significant gaps in our understanding of these systems, particularly in terms of their coordination and communication. Further research is needed to fully understand the potential and limitations of multi-agent LLM systems, as well as to develop effective evaluation frameworks and anti-collusion mechanisms.

## References

- [Understanding Multi-Agent LLM Frameworks: A Unified Benchmark and Experimental Analysis](#)
- [A Survey of Multi-Agent Deep Reinforcement Learning with Communication](#)
- [Towards Effective GenAI Multi-Agent Collaboration: Design and Evaluation for Enterprise Applications](#)
- [Cayley Graph Optimization for Scalable Multi-Agent Communication Topologies](#)
- [Interference-Aware K-Step Reachable Communication in Multi-Agent Reinforcement Learning](#)
- [AEMA: Verifiable Evaluation Framework for Trustworthy and Controlled Agentic LLM Systems](#)
- [Mapping Human Anti-collusion Mechanisms to Multi-agent AI Systems](#)
- [Augmenting the action space with conventions to improve multi-agent cooperation in Hanabi](#)
- [Towards Zero-Shot Coordination between Teams of Agents: The N-XPlay Framework](#)